

BC WILDWATCH™

CAMPUS ANIMAL SAFETY REPORTING ECOSYSTEM 3RD YEAR DEPLOYMENT PROJECT PLAN

Group 21

https://adolfjonker.github.io/Group_21_Wild_Watch_Project/

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1. INTRODUCTION

BC WildWatch™ is a campus animal safety reporting system designed for Belgium Campus iTversity, allowing students to report dangerous or nuisance animal sightings through a web-based Power Platform application. Originally developed as a second-year prototype, it has been upgraded into a fully deployed, scalable cloud solution integrating Power Apps, Power Automate, Power BI, Microsoft Entra ID authentication, and an AI chatbot. The project demonstrates the design, deployment, and management of a secure cloud application using Microsoft technologies while meeting professional software engineering and leadership standards.

2. TECHNOLOGY SELECTION JUSTIFICATION

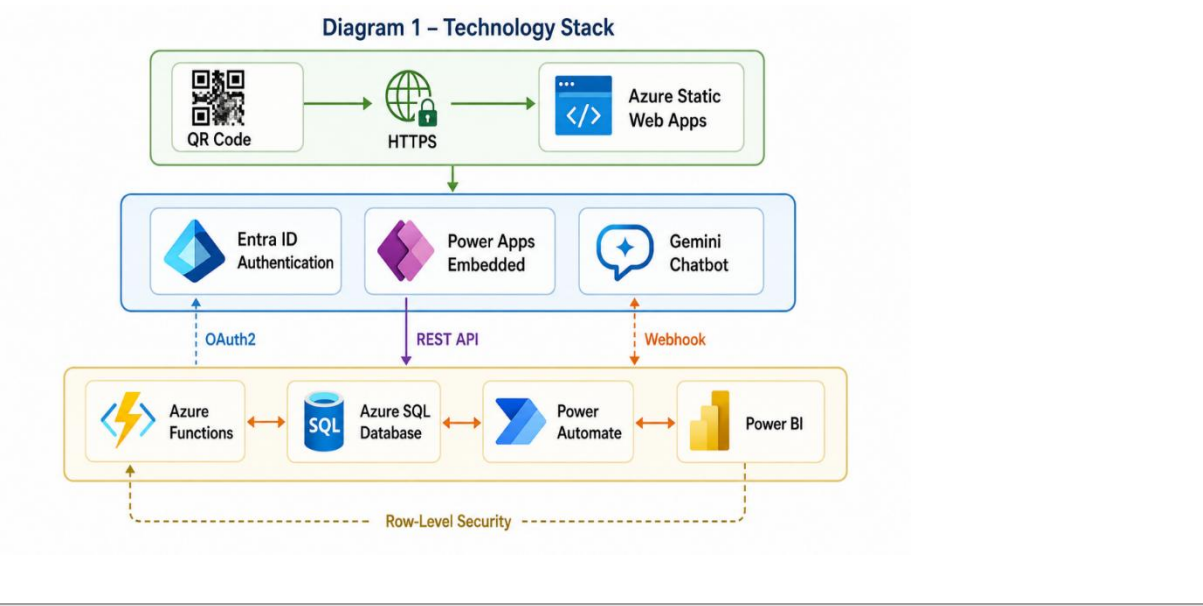
All technologies selected are **free-tier** and comply with project constraints.

Component	Technology Chosen	Justification
Frontend	Power Apps (Canvas App)	Low-code, native M365 integration, reuses Milestone 2
Web Hosting	Azure Static Web Apps (Free)	provides native GitHub integration with automatic CI/CD via GitHub Actions — every git push triggers a deployment, demonstrating professional DevOps practices.
Authentication	Microsoft Entra ID (A3)	Mandated SSO for @ belgiumcampus.ac.za accounts
Database	SharePoint Lists + SQL Azure (Free)	SharePoint for Power Apps writes; SQL for Power BI analytics
Automation	Power Automate	Triggers on SharePoint; sends Teams/Email alerts
Chatbot	Google Gemini API (Free Tier)	60 req/min free; better for African wildlife safety
Reporting	Power BI + Publish to Web (Free)	Live connection to SQL Azure; heatmap support
Version Control	GitHub (Public Repo)	Free, enables team collaboration, GitHub Actions

Component	Technology Chosen	Justification
QR Code	QR Code Monkey (Free)	Branded, high-res, trackable

3.1 Innovation Highlight

We implemented a **hybrid data layer**: SharePoint Lists handle real-time Power App writes, while Power Automate syncs data to SQL Azure for Power BI aggregations. This bypasses Dataverse licensing constraints while maintaining analytics performance. GitHub Actions CI/CD pipeline: Automated deployment on every commit to main branch, with deployment status visible in GitHub.



4. Deployment Strategy

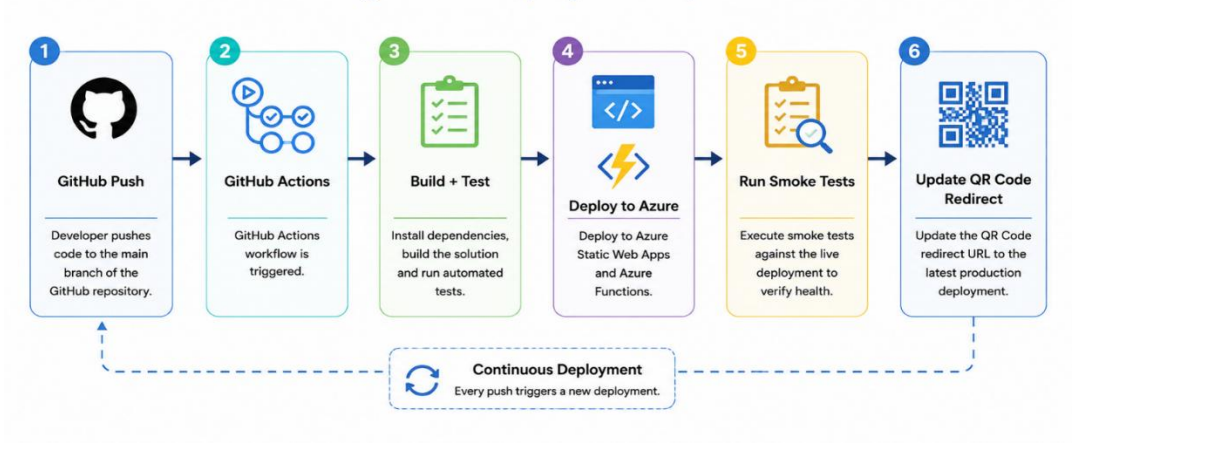
Phased deployment to ensure zero-downtime and rollback capability:

Phase	Activity	Timeline	Success Criteria
1	Local development with GitHub + Power Apps Studio	Week 1	All flows test in dev environment
2	Deploy Azure SQL schema + row-level security	Week 2	Security tests pass (users see only own reports)
3	Deploy Azure Functions + Power Automate webhooks	Week 3	Alerts trigger to Teams/email

BC WildWatch™ Deployment Project

Phase	Activity	Timeline	Success Criteria
4	Deploy Azure Static Web App + Entra ID auth	Week 4	Login works for all @belgiumcampus.ac.za
5	Embed Power Apps + Gemini chatbot	Week 5	Full user journey working
6	QR code generation + poster placement	Week 6	QR resolves to HTTPS URL
7	Load testing + security audit (OWASP ZAP free)	Week 7	No critical vulnerabilities
8	Go-live + demonstration	Week 8	Live demo successful

Diagram 2 – Deployment Pipeline



5. Risk Assessment and Mitigation

Risk	Likelihood	Impact	Mitigation Strategy
Azure free tier expires (12 months)	Medium	High	Document migration to Cloudflare Pages or Netlify
Entra ID auth fails for some emails	Low	High	Test with 10+ accounts; fallback manual report via Teams
Power Automate quota (2,000 requests/day)	Low	Medium	Batch sync every 15 min instead of real-time

Risk	Likelihood	Impact	Mitigation Strategy
Gemini API rate limit (60 req/min)	Medium	Medium	Cache common safety tips locally; implement retry logic
QR code poster vandalism	Low	Low	Generate 5 copies; store digital backup
Power BI web embed exposes raw data	Medium	High	Use Row-Level Security (RLS) with Entra ID; exclude student PII
SharePoint list lookup limits	Low	Medium	Use indexed columns; paginate Power Apps gallery

6. Data Security

The BC WildWatch™ solution incorporates multiple security controls to ensure the confidentiality, integrity, and availability of data throughout the system.

Data Encryption

- **Encryption at Rest:** All data stored within Azure SQL Database is protected using Transparent Data Encryption (TDE), which is included by default in the Azure free-tier environment.
- **Encryption in Transit:** Secure communication is enforced through TLS 1.3 across Azure Static Web Apps, Azure Functions, and Azure SQL Database, ensuring that all data transmitted between services remains encrypted.

Authentication and Identity Management

- User authentication is managed through **Microsoft Entra ID** using the OAuth 2.0 protocol with Proof Key for Code Exchange (PKCE). This approach enhances security by preventing unauthorized access and eliminating the need to store application passwords.

Access Control and Authorization

- **Student Users:** May create new incident reports and view only their own submissions. Access is restricted through Azure SQL Row-Level Security (RLS), using the authenticated user's principal name as the security filter.
- **Security Administrators:** Have permission to view, manage, and update all incident reports, as well as access analytical dashboards within Power BI.
- Access permissions are centrally managed through Microsoft Entra ID groups, which are mapped directly to Azure SQL database roles.

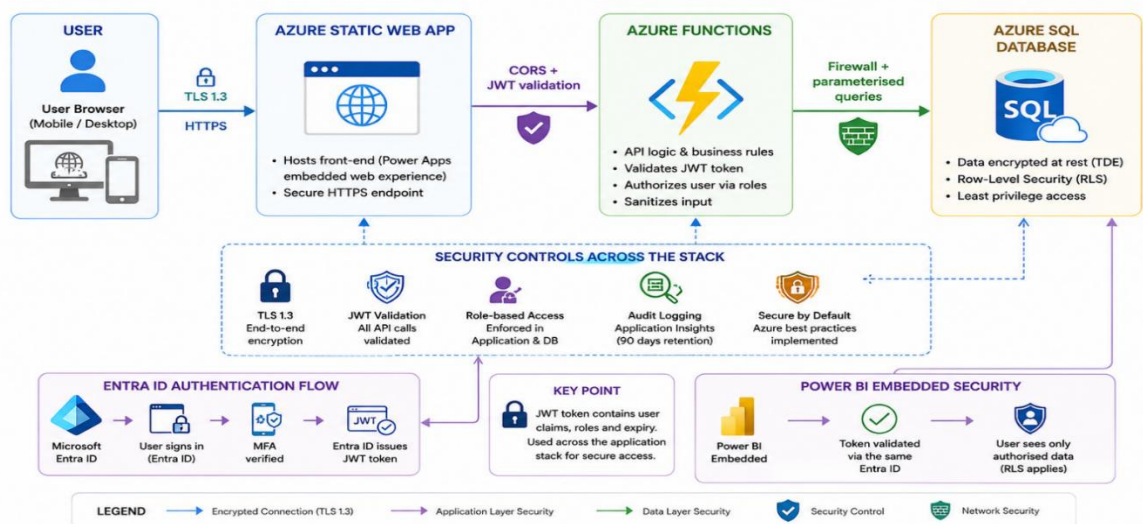
Monitoring and Audit Logging

- Application activity and operational events are captured through Azure Functions and stored in Azure Application Insights. Logging supports proactive monitoring, troubleshooting, and security auditing, with log retention configured for up to 90 days within the free-tier allocation.

Compliance and Privacy

- The solution is designed in accordance with Belgium Campus data protection requirements and South Africa's Protection of Personal Information Act (POPIA).
- Only essential information required for incident reporting is collected, and no unnecessary personally identifiable information (PII) is stored.
- Reports remain anonymous to the public, ensuring user privacy while still enabling effective incident management.

Diagram 3 – Security Architecture



7. Authentication and System Integration

Microsoft 365 A3 Authentication Integration

BC WildWatch™ integrates with **Microsoft Entra ID (formerly Azure Active Directory)**. A dedicated application registration was created with a secure callback URL linked to the Azure Static Web App. Access is restricted exclusively to @belgiumcampus.ac.za accounts. A custom student number attribute is included in the authentication process.

Power Apps Integration

The BC WildWatch™ user interface is developed using **Power Apps Canvas Apps** and embedded within the Azure Static Web App to provide a unified user experience. Communication between Power Apps and the backend services is handled through a secure custom connector that authenticates requests via Microsoft Entra ID.

When a user submits an animal sighting report, the information is transmitted through the Power Apps interface to Azure Functions, where business logic and validation are performed before the data is securely stored in Azure SQL Database. This architecture ensures efficient data processing while maintaining strong security controls.

Power Automate Integration

Power Automate is used to automate incident response workflows and improve communication between stakeholders. Whenever a new incident report is recorded in the database, an automated workflow is triggered to perform a series of actions.

These actions include:

- Sending an email notification to the campus security team containing incident details.
- Posting an adaptive card notification to the designated Microsoft Teams security channel for immediate visibility.
- Optionally sending an SMS notification through a third-party messaging service during demonstration scenarios.

This automation ensures that security personnel receive timely alerts and can respond rapidly to potential campus hazards.

Power BI Integration

To support data-driven decision-making, BC WildWatch™ incorporates **Power BI** dashboards connected directly to the Azure SQL Database. Data is imported and transformed into interactive visual reports that provide insights into animal sightings, incident trends, and high-risk locations on campus.

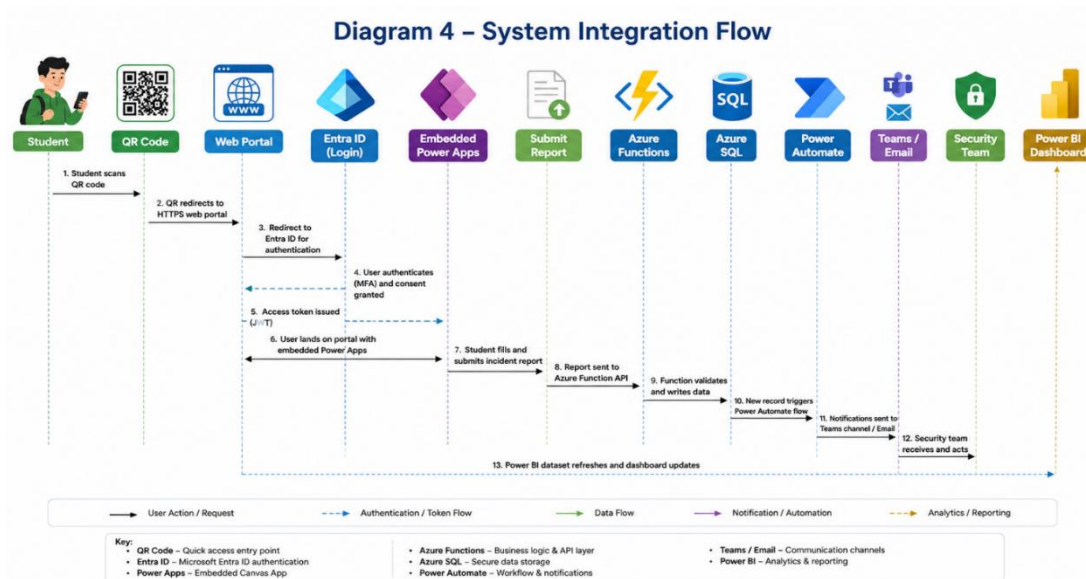
The dashboards are published through the Power BI Service and embedded within the web portal, allowing authorized users to access real-time analytics without leaving the application. Access to reporting data is governed through the same Microsoft Entra ID authentication framework used throughout the platform.

Chatbot Integration

The platform also includes an AI-powered chatbot designed to assist users with animal safety information and reporting guidance. The chatbot communicates through Azure Functions, which securely interact with the AI service and return responses to the Power Apps interface.

The chatbot is configured to provide safety recommendations for common campus wildlife hazards such as snakes, bees, and stray animals. To ensure responsible usage, responses are restricted to general safety guidance and do not provide medical or emergency advice. Frequently requested information is cached to improve performance and minimize operational costs.

By integrating Microsoft Entra ID, Power Apps, Power Automate, Power BI, and AI-assisted support services, BC WildWatch™ delivers a secure, scalable, and highly connected solution that meets both the technical and operational requirements of Belgium Campus.



8. QR Code Implementation Plan

Requirements Fulfilled

- The QR code redirects users directly to the **public HTTPS Azure Static Web App URL**.
- It is generated using **free tools and APIs**, with optional branding applied.
- The QR code is displayed during the **live demonstration session**.
- It is distributed across **posters, notice boards, mobile devices, and student CV portfolios** to maximise accessibility and visibility.

QR Code Generation Methods

Two generation approaches were implemented to ensure reliability and flexibility:

Method	Tool Used	Output Format	Branding
Primary	QR Server API (api.qrserver.com)	PNG (500×500)	Logo overlay via ImageMagick (free tool)
Secondary	Canva QR Generator (free version)	PNG + SVG	Full BC WildWatch™ branded design

Technical Implementation

```
curl "https://api.qrserver.com/v1/create-qr-code/?size=300x300&data=https://[app].azurestaticapps.net" > qr.png
```

```
composite -gravity center -geometry 70x70 logo.png qr.png branded-qr.png
```

Fallback Strategy

If the QR code is unavailable or damaged, users can still access the system using a shortened fallback URL:
[https://adolfonker.github.io/Group_21_Wild_Watch_Project/Tracking and Analytics](https://adolfonker.github.io/Group_21_Wild_Watch_Project/Tracking_and_Analytics)

QR scan analytics are monitored using a free Bitly account, which provides insights such as:

- Number of scans

- Device type (mobile/desktop)
- Geographic location of scans

These analytics are demonstrated during the live presentation to showcase system engagement.

Placement Strategy (Demonstration Day): To ensure maximum accessibility and visibility, the QR code will be placed in multiple locations:

- Printed on an A4 poster next to the demonstration laptop
- Included in student CVs under the portfolio section.
- Embedded in presentation slides for live scanning during the demo.

8.1. BC WildWatch™ Branded QR Code

https://adolfjonker.github.io/Group_21_Wild_Watch_Project/



9. RACI Responsibility Table

RACI Legend:

R = Responsible (does the work)

A = Accountable (answers for success/failure – only one per task)

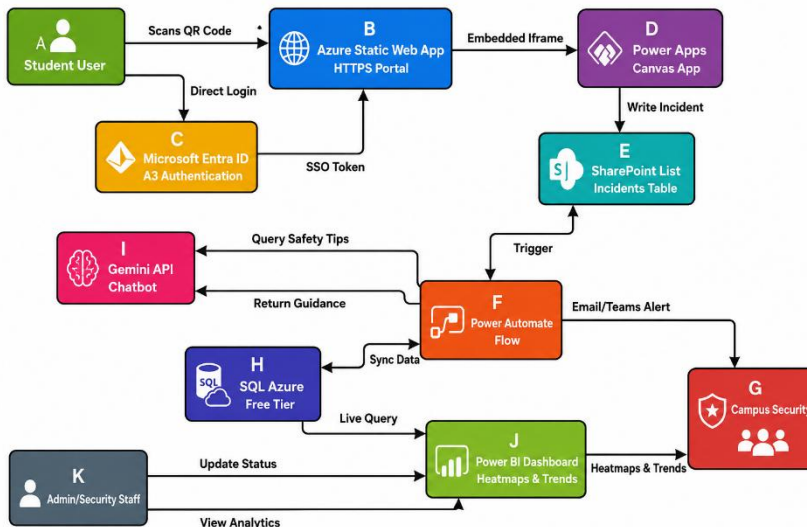
C = Consulted (two-way communication)

I = Informed (one-way updates)

BC WildWatch™ Deployment Project

Task	PM	Apps	Backend	UI/UX	BI	Docs	Deploy	QA	Security	AI
Project Planning	RA	C	C	C	C	C	C	C	C	C
Azure Deployment	C	I	I	I	I	I	RA	C	C	I
Entra ID Authentication	C	I	I	I	I	I	C	I	RA	I
Power Apps Embed	A	RA	C	C	I	I	C	I	I	I
SharePoint + SQL	C	C	RA	I	C	I	C	C	I	I
Power Automate Alerts	A	I	RA	I	I	I	C	C	C	I
Power BI Dashboard	A	I	C	I	RA	I	I	C	C	I
QR Code + Branding	A	I	I	RA	I	I	I	I	I	I
Testing & Validation	A	C	C	C	C	C	C	RA	C	C
Video Presentation	RA	I	I	I	I	C	I	I	I	C

10 Architecture Diagram



11. Live Deployment Demonstration Plan

Time	Activity	Screens Showing
0:00	Scan QR code with mobile phone	Phone camera → browser opens
0:30	Entra ID login	Login page → @ belgiumcampus.ac.za
1:00	Submit incident (snake near Building A)	Power Apps form with photo upload
2:00	Show SharePoint new item	SharePoint list → new row appears
2:30	Power Automate trigger	Teams message to Security channel
3:00	Chatbot interaction	"What if bitten by snake?" → Gemini response
4:00	Security admin view	Admin app → update status to "Resolved"
5:00	Power BI dashboard	Heatmap shows Building A high-risk
6:00	Student "My Reports" gallery	Status shows "Resolved"
7:00	Q&A	Open floor